Cartesian closed categories of domains

Lean 4 Formalization Companion – Chapter 4

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Section 4.0

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Definition 4.0.1. Let D and E be dcpos, with E pointed. For $d \in D$ and $e \in E$, the step function $(d \searrow e) : D \to E$ is

$$(d \searrow e)(x) = \begin{cases} e, & x \in \mathrm{Int}(\uparrow d), \\ \bot_E, & \text{otherwise.} \end{cases}$$

More generally, $(O \searrow e)$ maps the Scott-open set O to e and maps its complement to $\bot_E$.

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Proposition 4.0.2.

1. Step functions are Scott-continuous.
2. If D, E are dcpos, E is pointed, $f : D \to E$ is continuous, and $e \ll f(d)$, then $(d \searrow e) \ll f$.
3. If, in addition, D and E are continuous, then every continuous $f : D \to E$ is a supremum of step functions.

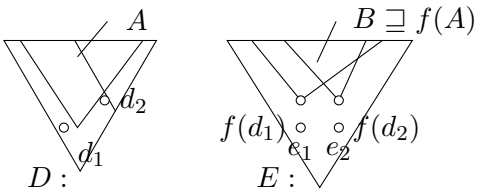


Figure 10: Finding an upper bound for two step functions.

Section 4.1

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Definition 4.1.1. Let E be a pointed continuous domain. Lean

1. E is an L-domain if every pair $e_1, e_2 \leq e$ has a supremum in $\downarrow e$.
2. E is bounded-complete if every bounded pair has a supremum.
3. E is a continuous lattice if every pair has a supremum.

The corresponding full subcategories are denoted L, BC, and LAT, with algebraic counterparts aL, aBC, and $aLAT$.

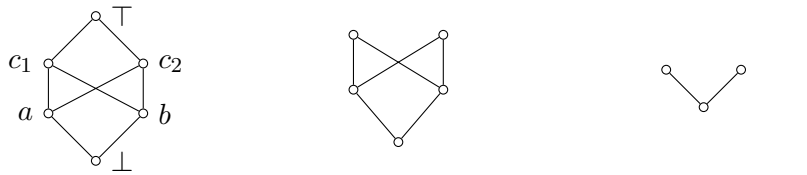


Figure 11: Separating examples for the categories of lattice-like domains.

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Proposition 4.1.2. For a pointed continuous domain D, being an L-domain, a bc-domain, or a continuous lattice is equivalent respectively to having infima for bounded nonempty subsets, nonempty subsets, or arbitrary subsets. Lean

Proposition 4.1.3. Retracts and bilimits of L-domains, bc-domains, and continuous lattices are again objects of the same respective class. Lean

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Proposition 4.1.4. If D is an L-domain, a bc-domain, or a continuous lattice, then $\mathrm{Idl}(D, \sqsubseteq)$ is respectively an algebraic L-domain, an algebraic bc-domain, or an algebraic lattice. Lean

Proposition 4.1.5. If D is a continuous domain and E is an L-domain, a bc-domain, or a continuous lattice, then the function space $[D \to E]$ is again respectively an L-domain, a bc-domain, or a continuous lattice. Lean

Corollary 4.1.6. The categories L, BC, LAT, and their algebraic counterparts are cartesian closed. Lean

Section 4.2

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Definition 4.2.1. Let P be a poset.

1. P is mub-complete if every upper bound of a finite set $M \subseteq P$ lies above a minimal upper bound of M; equivalently,
$$\bigcap_{m \in M} \uparrow m = \uparrow \operatorname{mub}(M).$$
2. The mub-closure $\operatorname{mc}(A)$ is the least superset of A closed under adjoining $\operatorname{mub}(M)$ for finite M.
3. P has the finite mub property if it is mub-complete and every finite subset has finite mub-closure; with a least element it is a Plotkin order.
4. An algebraic domain whose compact basis is a Plotkin order is called bifinite; the full subcategory of $ALG_\perp$ is denoted B.

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Definition 4.2.2. Let D be bifinite and E pointed algebraic. A finite set $F \subseteq K(D) \times K(E)$ is joinable if
$$\forall G \subseteq F\ \exists H \subseteq F\ \Big(\pi_1(H) = \operatorname{mub}(\pi_1(G)) \ \wedge\ \forall c \in \pi_2(G)\ \forall d \in \pi_2(H),\ c \sqsubseteq d\Big).$$
The function associated with a joinable family F is
$$x \longmapsto \bigvee \{e \mid \exists d \in K(D),\ d \sqsubseteq x \ \wedge\ (d,e) \in F\}.$$

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Lemma 4.2.3. If D is bifinite and E is pointed algebraic, every joinable subset of $K(D) \times K(E)$ gives a compact element of $[D \to E]$. For joinable families F, G, the associated functions satisfy the stated order exactly when every $(d,e) \in G$ is matched by some $(d',e') \in F$ with $d' \sqsubseteq d$ and $e \sqsubseteq e'$.

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Proposition 4.2.4. If D is bifinite and E is pointed algebraic, then $[D \to E]$ is algebraic, and all compact elements of $[D \to E]$ arise from joinable families. *Proof sketch.* Let $f : D \to E$ be continuous, let M be a finite mub-closed set of compact elements of D, and choose compact elements $(e_m)_{m \in M}$ of E with $e_m \sqsubseteq f(m)$. One constructs compact elements $(\widehat{e}_m)_{m \in M}$ so that $m \mapsto \widehat{e}_m$ is order-preserving, by repeatedly choosing an upper bound for $\{e_m\} \cup \{\widehat{e}_{m'} \mid m' < m\}$. This yields a directed collection of compact elements of $[D \to E]$ arbitrarily close to f.

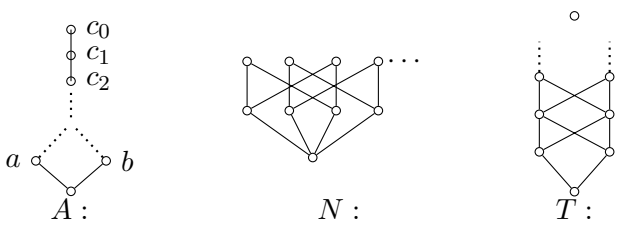


Figure 12: Typical non-bifinite domains.

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Proposition 4.2.5. Let D be algebraic with mub-complete basis $K(D)$, and let $A \subseteq K(D)$. There is a least kernel operator p_A on D fixing A, given by

$$p_A(x) = \bigvee \{c \in \mathrm{mc}(A) \mid c \sqsubseteq x\}.$$

Theorem 4.2.6. For a pointed dcpo D, the following are equivalent: Lean

1. D is bifinite.
2. There is a directed family $(f_i)_{i \in I}$ of idempotent deflations with $\bigvee_i f_i = \mathrm{id}_D$.
3. The set of all idempotent deflations is directed and has join id_D.

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Theorem 4.2.7. A dcpo is bifinite if and only if it is a bilimit of an expanding system of finite pointed posets. *Proof sketch.* If D is bifinite, take the images of a directed family of idempotent deflations as finite pointed posets, with connecting embeddings induced by inclusion of images; the upper adjoint is the corresponding deflation restricted to the image. Conversely, if D is a bilimit of finite posets, the composites $f_i \circ g_i$, where g_i is the upper adjoint of the embedding f_i, give the directed idempotent deflations required in Theorem 4.2.6. Lean

Theorem 4.2.8. The category B of bifinite domains is closed under cartesian products, function spaces, coalesced sums, and bilimits. In particular, B is cartesian closed. *Proof sketch.* For function spaces, if $(f_i)_{i \in I}$ and $(g_j)_{j \in J}$ are directed families of idempotent deflations for D and E, then Lean

$$F_{ij}(h) = g_j \circ h \circ f_i \qquad (i, j) \in I \times J$$

forms a directed family of idempotent deflations on $[D \to E]$.

Definition 4.2.9. A Scott-continuous $f : D \to D$ is finitely separated from id_D if some finite $M \subseteq D$ satisfies: for every $x \in D$, there is $m \in M$ with $f(x) \sqsubseteq m \sqsubseteq x$. Strong separation requires $m, m' \in M$ with $f(x) \sqsubseteq m \ll m' \sqsubseteq x$. A pointed dcpo is an FS-domain if a directed family of finitely separated continuous self-maps has supremum id_D. Lean

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Proposition 4.2.10 (conjectural in the source). If D is an FS-domain and E is pointed continuous, then $[D \to E]$ is continuous. The immediately following comment says that the proof was as yet unknown. The Lean side therefore records only the statement and does not provide evidence for it. Lean

Theorem 4.2.11. The category FS is closed under products, function spaces, coalesced sums, and bilimits; hence it is cartesian closed. Lean

Proposition 4.2.12. Lean

1. Every bifinite domain is an FS-domain.
2. A retract of an FS-domain is an FS-domain.
3. An algebraic FS-domain is bifinite.

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Theorem 4.2.13. For a continuous domain D, the sets of open neighborhoods of compact saturated sets are exactly the Scott-open filters in σD. Lean

Proposition 4.2.14. For a continuous domain D: Lean

1. κD is a dcpo, with directed suprema given by intersection.
2. If a filtered intersection of compact saturated sets is contained in a Scott-open set O, then one member is already contained in O.
3. $\kappa D \setminus \{\emptyset\}$ is a dcpo.

Proposition 4.2.15. For a continuous domain D: Lean

1. κD is continuous.
2. $A \ll B$ in κD iff there is a Scott-open O with $B \subseteq O \subseteq A$.
3. $O \ll U$ in σD iff there is compact saturated A with $O \subseteq A \subseteq U$.

The proof uses upper sets generated by finitely many points: if O is an open neighborhood of a compact saturated set A, then there is a finite set $M \subseteq O$ with

$$A \subseteq\uparrow M \subseteq\uparrow M \subseteq O.$$

Proposition 4.2.16. A continuous domain D is coherent iff for all $O, U_1, U_2 \in \sigma D$, from $O \ll U_1$ and $O \ll U_2$ it follows that $O \ll U_1 \cap U_2$. Lean

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Proposition 4.2.17. An algebraic domain D is coherent iff $K(D)$ is mub-complete and finite subsets of $K(D)$ have finite sets of minimal upper bounds. This is the "2/3-SFP Theorem" of Plotkin: coherence rules out the first two non-bifinite examples from Figure 12, but not the third. In algebraic coherent domains, the compact-open sets $\uparrow c_1 \cup \cdots \cup \uparrow c_n$ form a common sublattice of σD and κD. Lean

Theorem 4.2.18. FS-domains and bifinite domains are coherent. Lean

Definition 4.2.19. The Lawson topology on a dcpo D is the smallest topology containing all Scott-open sets and all complements $D\setminus\uparrow x$. It is denoted by λD. Lean

Proposition 4.2.20. For a continuous domain D: Lean

1. The Lawson topology is Hausdorff, and every Lawson-open has the form $O \setminus A$ with O Scott-open and A Scott-compact saturated.
2. The Lawson topology is compact iff D is coherent.
3. Scott-continuous retracts of Lawson-compact continuous domains are Lawson-compact and continuous.

Section 4.3

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Lemma 4.3.1. If D, E are continuous domains, E is pointed, and $[D \to E]$ is continuous, then D is coherent or E is an L-domain. *Proof sketch.* Suppose not. Choose Scott opens O, U_1, U_2 with $O \ll U_1$, $O \ll U_2$, but not $O \ll U_1 \cap U_2$, and a directed cover $(V_i)_{i \in I}$ of $U_1 \cap U_2$ none of whose members covers O. Interpolate $O \ll U_1' \ll U_1$ and $O \ll U_2' \ll U_2$. The failure of the L-domain property reduces to two retract cases. For $E = A$, take $f_1 = (U_1' \searrow a)$, $f_2 = (U_2' \searrow b)$, and

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$$f(x) = \begin{cases} c_0, & x \in U_1 \cap U_2, \\ a, & x \in U_1 \setminus U_2, \\ b, & x \in U_2 \setminus U_1, \\ \bot, & \text{otherwise.} \end{cases}$$

Given an upper bound g of $\{f_1, f_2\}$ below f, define

$$h_i(x) = \begin{cases} c_0, & x \in V_i, \\ c_{n+1}, & x \in (U_1 \cap U_2) \setminus V_i \text{ and } g(x) = c_n, \\ g(x), & \text{otherwise.} \end{cases}$$

No h_i is above g, but their supremum is f, contradicting continuity of $[D \to E]$. For $E = X$, use the same opens and

$$f(x) = \begin{cases} c_1, & x \in U_1 \cap U_2, \\ a, & x \in U_1 \setminus U_2, \\ b, & x \in U_2 \setminus U_1, \\ \bot, & \text{otherwise,} \end{cases} \qquad h_i(x) = \begin{cases} \top, & x \in V_i, \\ c_2, & x \in (U_1 \cap U_2) \setminus V_i, \\ g(x), & \text{otherwise.} \end{cases}$$

to obtain the same contradiction.

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Lemma 4.3.2. A continuous domain with bottom is an FS-domain iff both D and $[D \to D]$ are coherent.

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Theorem 4.3.3. Every cartesian closed full subcategory of $CONT_\bot$ is contained in FS or in L.

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Theorem 4.3.4. Every cartesian closed full subcategory of $ALG_\bot$ is contained in B or in aL.

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Theorem 4.3.5. Every cartesian closed full subcategory of ω-$CONT_\bot$ respectively ω-$ALG_\bot$ is contained in ω-FS respectively ω-B.

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Lemma 4.3.6. If D, E are continuous domains and $[D \to E]$ is continuous, then D is compact or E is a disjoint union of pointed domains.

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Definition 4.3.7. A dcpo D is a finite amalgam if it is the union of finitely many pointed dcpos $D_1, \ldots, D_n$ such that every intersection of members of this family is also a union of members of the family. Categories of finite amalgams of objects of C are denoted F-C, and disjoint unions by U-C.

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Proposition 4.3.8. A mub-complete dcpo is a finite amalgam iff the mub-closure of the empty set is finite.

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Lemma 4.3.9. If both D and $[D \to D]$ are compact and continuous, then D is a finite amalgam. Lean

Theorem 4.3.10. Lean

1. The maximal cartesian closed full subcategories of $CONT$ are F-FS, U-FS, F-L, and U-L.

2. The maximal cartesian closed full subcategories of ALG are F-B, U-B, F-aL, and U-aL.